

SCENARIO 4 LAB BOOK

Monday 12th December

Movements of the Arm Relative to the Cursor

For Click

Anterior arm muscle- Flexor Carpi Radialis

For Leftward Movement

Anterior arm muscle- Flexor Carpi Ulnaris

For Rightward Movement

Brachioradialis

EMG background

Electromyography (EMG) is a diagnostic procedure to assess the health of our muscles and the nerve cells that control them which are known as the motor neurons. EMG results can reveal nerve dysfunction, muscle dysfunction and even problems with nerve-to-muscle signal transmission.

Motor neurons transmit electrical signals that cause muscles to contract. An EMG uses electrodes to translate these signals into graphs hence when you contract the muscles attached to the electrodes these electrical signals translate onto the graph. These muscle contractions are compared to a reference EMG of ideal healthy muscles.

The part we are focused on is a nerve conduction study, it is another part of an EMG, uses electrode stickers applied to the skin (surface electrodes) to measure the speed and strength of signals traveling between two or more points marked with the electrode.

[Electromyography \(EMG\) - Mayo Clinic](#)

“+- 500 to +-600” microvolts are the main/ dominant voltage range of the EMG recorded for muscles

Reference: [Frequency spectrum of EMG signal | Download Scientific Diagram \(researchgate.net\)](#)

The 0-5 volts is the Arduino power range as we decided to use the Arduino Leonardo to power our EMG.

Let's say we want to display it at roughly 3 volts instead of 5 volts to reduce the chances of saturation which could occur near the top as this could potentially cut off the signal.

For an EMG signal 600 micro volts:

$$3 \div (600 \times 10^{-6}) = 5000 \div 600 \times 10^{-6} = 5000$$

, hence, we would initially like use 5000 as our overall gain for the maximum output value of 3 volts using the Arduino power supply.

For an EMG signal 500 micro volts:

$$5000 \times (500 \times 10^{-6}) = 2.55000 \times 500 \times 10^{-6} = 2.5$$

volts hence this will be the minimum expected value on the Arduino power supply. Or the digital levels on the serial monitor that correspond to these voltage levels.

Therefore, initially aiming for a gain of 5000 is what we are aiming for in order to get the voltage range of 2.5-3 volts, so it is hence within the range of the Arduino power supply.

Using instrumentation amplifier and a non-inverting amplifier to amplify the EMG with a total gain of around 5000 was our initial idea however we realized this gain may not be successfully reached as all as the voltage input for EMG is extremely small hence the op-amp won't act like an ideal op-amp as it is designed for higher voltages hence values will need to be changed along the way.

DC offset caused by a DC signal moves the signal away from the base/center line hence this signal is: $\pm 600 \text{ micro volts} + \text{the DC offset} = \text{the signal}$

Gain of around 10 for the instrumentation amplifier because if we amplify the entire signal with the instrumentation amplifier the DC offset will also get amplified, this will amplify the unwanted parts of the signal hence form an amplification that is too high so the signal will become saturated as it's beyond the Arduino voltage range. Hence, we only amplify slightly here so that the small signal is clear enough to be seen and then we undergo filtering to cut-off the DC offset and noise) before undergoing the final amplification at the non-inverting amplifier to get the signal into the Arduino range.

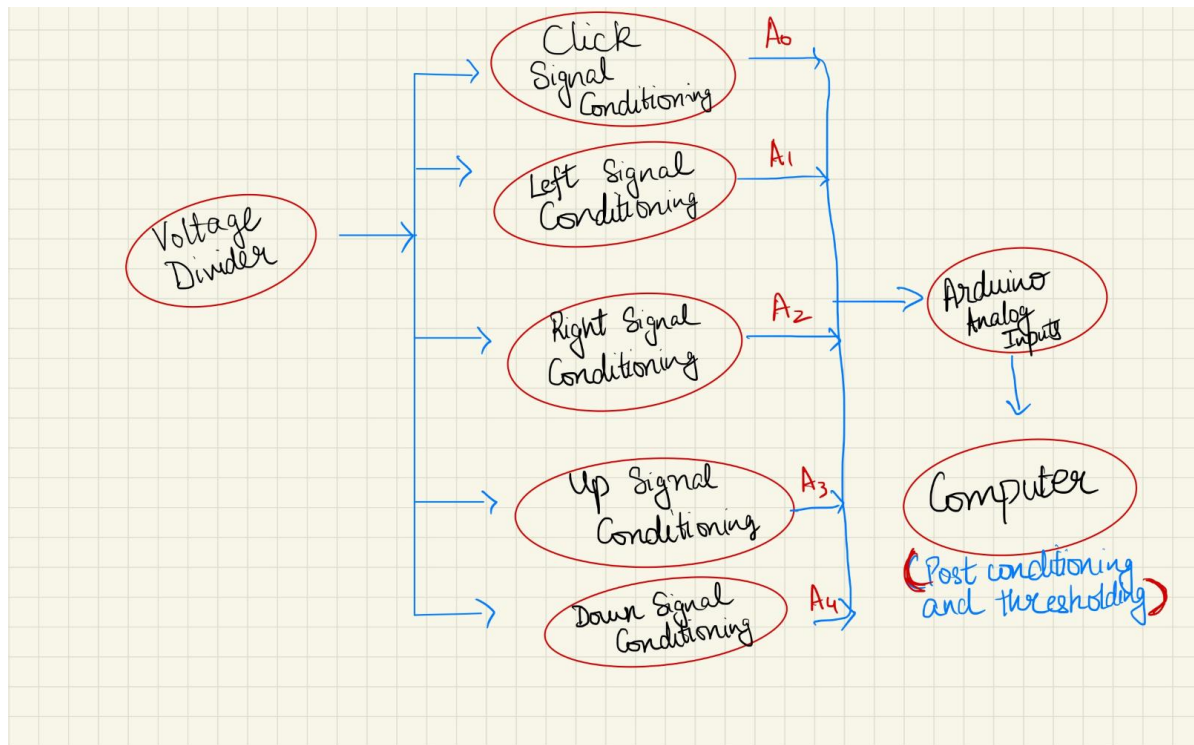
This leaves a Gain of 5000 for non-inverting amplifier: $5000/10 = 500$

The dominant Frequency range for muscles is “20-150” Hz. The general EMG range lies between “6-500Hz”, but the dominant cut-off frequency narrows the range to the more expected and common frequency range to help cut-off more noise and reduce it to the most common and desired frequencies as a narrower range the better. Initially the cut-off frequencies for the high pass filter is 20 Hz and the active 2nd order low pass filter it is 150 Hz.

Reference: [\(PDF\) Electromyography Signal on Biceps Muscle in Time Domain Analysis \(researchgate.net\)](#)

[Signal conditioning for the 4 muscles and the click](#)

[Overall flow chart](#)



Signal conditioning of the circuit to form an EMG

We first decided to build a circuit to form an EMG in order to test for the click. The goal was every time a surface muscle clenches, a click is then produced, and this is programmed using the Arduino software. It is the difference in electric potential between the two points in the muscle during the clench that forms this click. In order to test it we first needed to design a signal conditioning circuit for it with the correct frequency range and voltages. We decided to make the muscle conditioning circuit used for the click also convenient for other muscle uses in terms of frequencies so that the design could be repeated for each muscle circuit as well hence we did make the EMG range we decided to use slightly broader now from 20-500 Hz to make the circuit design also work for the other muscles if we design it for a broader frequency range to save time and effort. These frequencies are still the correct range for an EMG, it is just the less dominant range as 500 Hz is the maximum value to use within a common EMG range.

1-voltage divider circuit and buffer

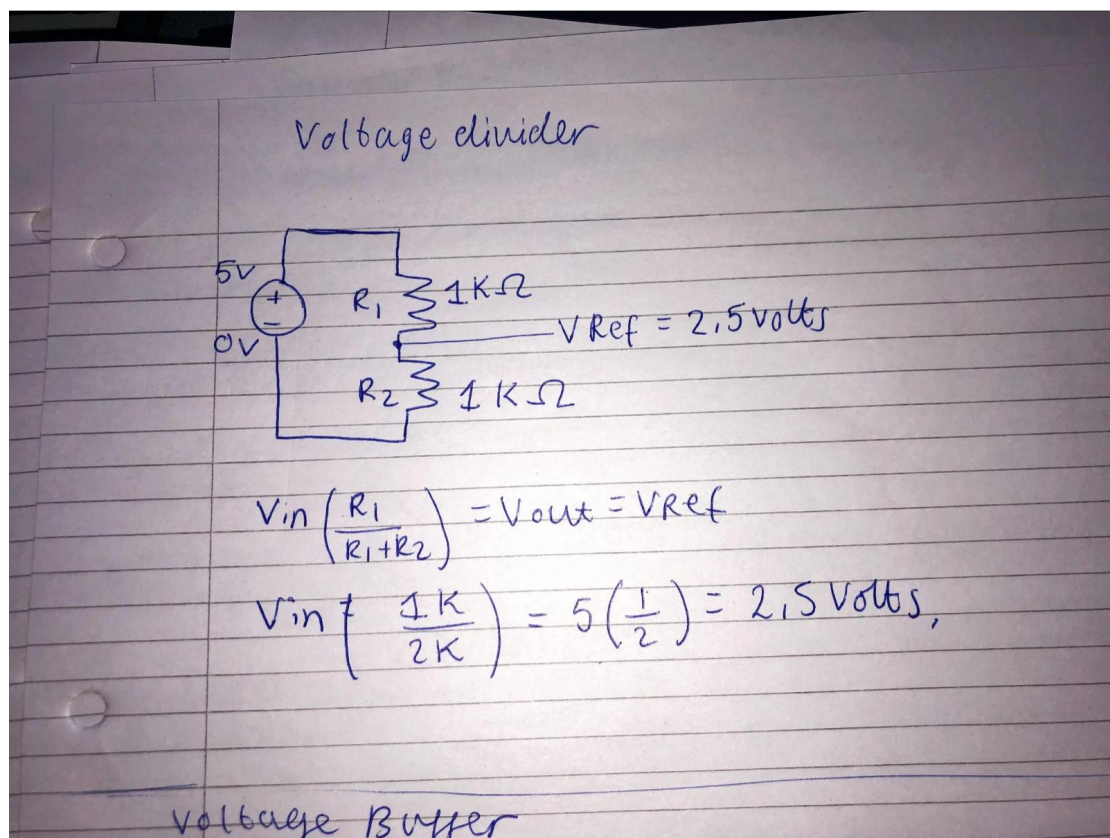
We are powering our circuit from the Arduino and the Arduino does not supply negative volts hence does not supply between 0 to -5 volts therefore our voltage reference no longer becomes 0 as our new voltage range is 0-5 volts. Our new middle value is more convenient to make as our voltage reference hence 2.5 volts was used as our V_{ref} that acts like a ground. Therefore, once the signal has been amplified and placed in the Arduino range, the negative EMG values will not be cut-off as the range has shifted upwards into the positive range as V_{ref} is now 2.5 volts which acts as a center/base line instead of V_{ref} being at zero volts, hence the negative parts of the EMG signal does not get cut-off.

This meant for now we did not need a rectifier circuit as all the negative values did not have to be cut off as they were shifted upwards into the positive voltage range where the signal lies around the Centre line V_{ref} at 2.5 volts. And the amplification

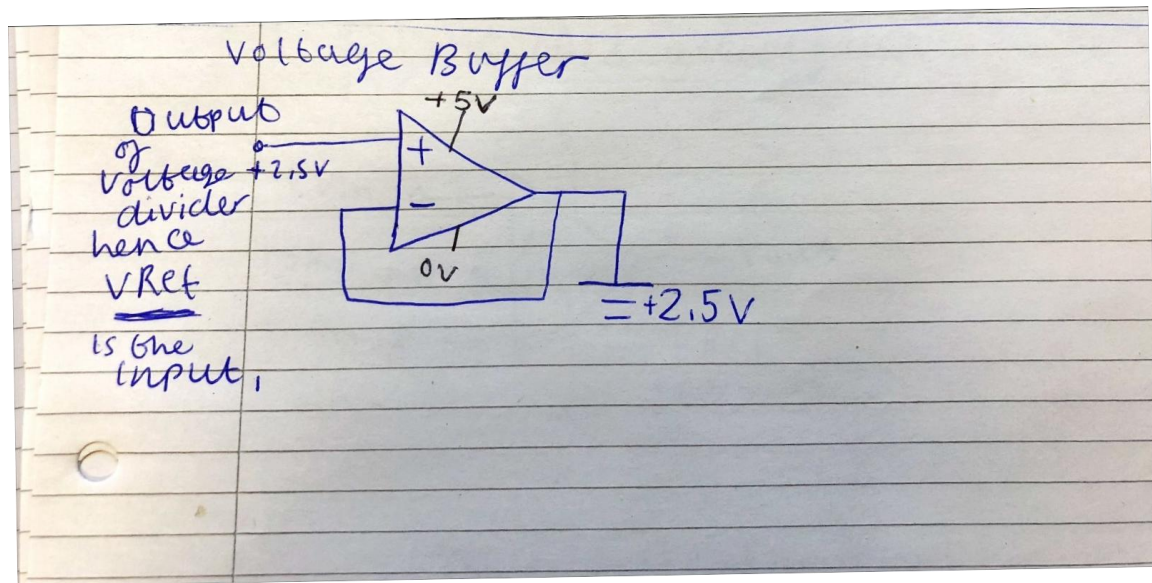
of our circuits was built to fit into the Arduino range. The rectifier converts the alternating current to direct current hence all the negative part of the signal is shifted to the positive so only allows current to flow in one direction reversing the current this was not needed as our V ref shifted the signal upwards hence the entire signal is already positive when using a v ref of 2.5 volts

In order to set V ref to 2.5 volts a voltage divider circuit was used between 0-5 volts, when two identical resistors are placed in series a voltage divider circuit forms as the overall voltage are split equally between the two identical resistors as they share an equal portion of the overall voltage making the point between the two resistors $5/2 = 2.5$ volts, this point is our reference point on the circuit.

Voltage divider circuit:



However, before we connected it to an instrumentation amplifier a voltage buffer circuit was needed in order to stop loading as the instrumentation amplifier would load the voltage divider circuit hence change the output of the voltage divider circuit, in order to stop this a voltage buffer circuit was placed in-between the two. A voltage buffer is a non-inverting amplifier where the input impedance is infinite in order to limit the current being drawn out the circuit hence the previous circuit will not load/change its behavior as a result of later circuits.



2-Instrumentation amplifier

An instrumentation amplifier was needed in order to take the voltage difference across the 2 input signals at the two electrodes (one at each electrode) placed on the muscle and then amplify the difference between them. It takes the EMG recording then amplifies it as the EMG tends to be a very small signal hence undergoes some of the amplification to make the signal more visible. An instrumentation amplifier was chosen over a differential amplifier as it means only one variable resistor R_G needs to be changed in order to adjust the gain leaving the other resistors constant hence it is easy and convenient in order to change the gain. When changing the gain of differential amplifier, you also slightly change the other resistance values. This will lead to incorrect amplified version of the difference and hence you will be getting distortion.

Also, it's because the input impedance of the instrumentation amplifier is almost infinite hence less / almost no current is drawn out of the circuit, a lot of current being drawn out can lead to future problems such as when connection the Arduino Leonardo the Arduino software is unable to recognize this device, so we want to limit the amount of current being drawn out.

The differential amplifier also has a higher common mode rejection ratio making it behave more like an ideal op-amp, so we are more likely to get expected results. The instrumentation amplifier used was the “INA 126”

We initially decided to use a gain of 10 which seemed correct due to the math's however we tested the gain of an instrumentation amplifier by inputting a signal from the signal generator with an input voltage in the microvolts range however the increase in gain was extremely small hence we decided it was not going to be a big enough amplification which makes sense because even though the math's was correct, the op-amp will not undergo its ideal behavior as the signal is so small (within the microvolts range). we decided to use a gain of 20 instead as our amplification was overall needing a gain of around 5000 whereas ECG was around 1000 when the instrumentation amplifier gain was 10, hence we thought a good value to try out would be 20 for our instrumentation amplifier for an EMG as it is a signal smaller than the ECG range so will need more amplification at this point. From the data sheet we found that the gain was:

$$[Equation] = 20 \text{ IDEAL GAIN}$$

$$RG = 5333 \text{ Ohms}$$

$$RG = 4.7 \text{ K ohms}$$

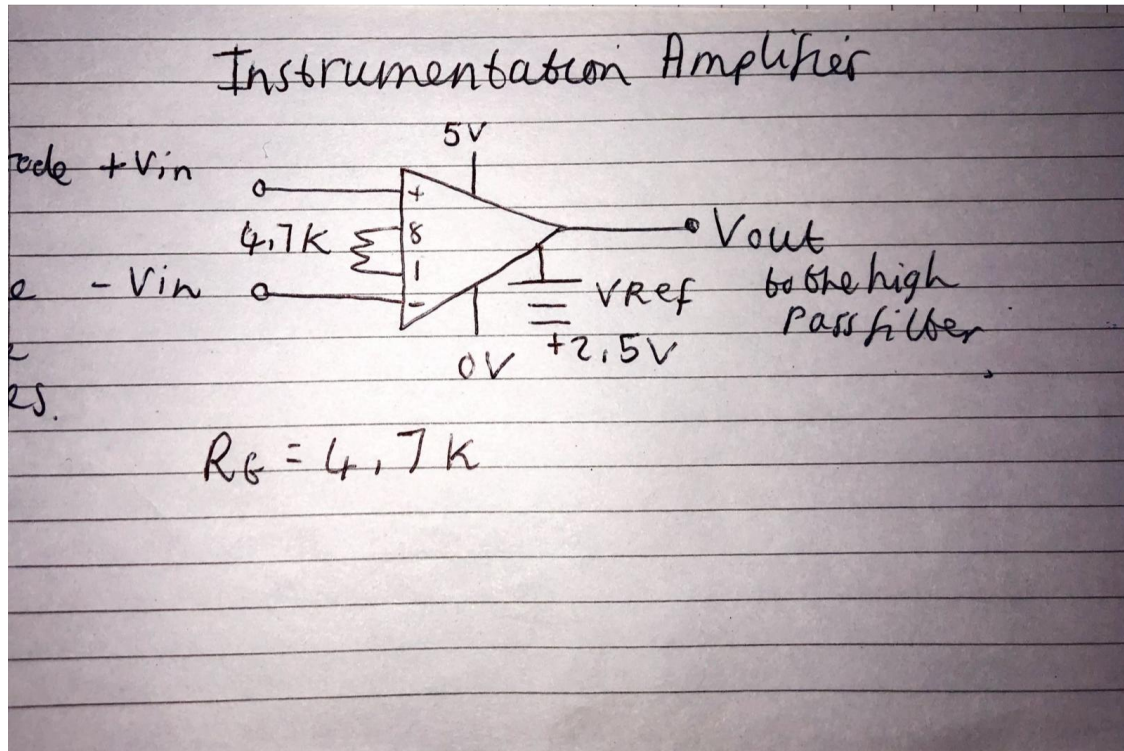
Hence our actual gain is:

$$[Equation]$$

To summarize:

So the gain =22

RG= 4.7K ohms



Here the two inputs at the input terminal would be the positive and negative electrodes, this does not matter with muscles however if they are reversed you will just get an inverting signal and therefore can just switch them around.

So our actual gain is 22 for the instrumentation amplifier we decided it's okay to have a higher gain as the voltage is very small so whatever happens in our op-amp is ideally towards higher voltages not very small voltages such as millivolts so our calculated gain will not correspond with our actual output gain hence, we estimated that this gain would be better if it is higher anyway, so we went with this to start off with as a lot more amplification was needed for microvolts range.

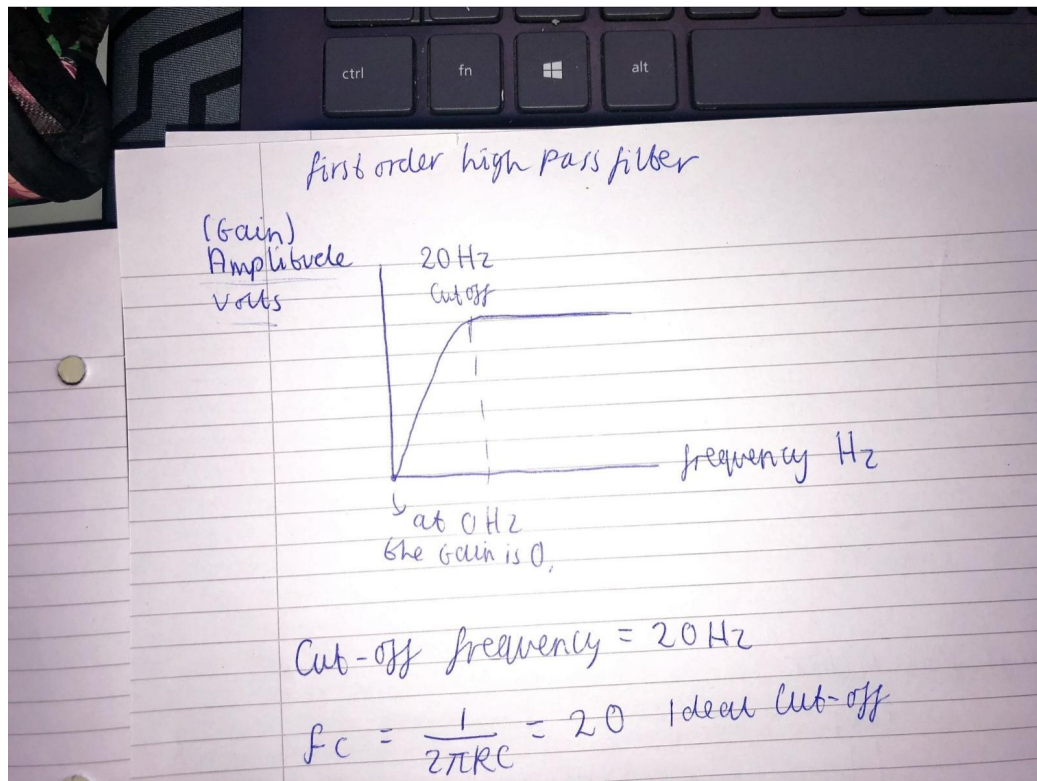
The Voltage input terminals will be from the two electrodes placed accurately on a surface muscle, one a positive electrode and one negative electrode, placed closely together in order to get a clearer EMG signal, along with the V reference electrode placed on a bony area like the elbow due to the elbow having the least amount of muscle activity during flexion and extension hence will remain unaffected by muscle movement and won't have an effect on the EMG hence is the body part used as a reference as it doesn't impact the results.. This will input a signal into the

instrumentation amplifier, this signal measures the difference in electrical potential between the two muscles in order to form an EMG.

Tuesday 13th December

First order high pass filter

This filter is specifically used in order to cut off the 0 Hz DC offset that was mentioned earlier. By cutting this off the signal won't have an offset added onto it which could lead to saturation in the final signal/ some of the signal will be lost as it will unlikely fit in the range. This offset occurs at 0 Hz hence a high pass filter is used to cut off this low/ no frequency. However as the smallest value in our preferred range was 20 Hz it's easier to set this value as our cut-off as we might Aswell cut out the unnecessary frequencies, a first order high pass filter was used instead of second order as don't need to exactly and instantly cut off all frequencies below 20 Hz we only mainly need to remove the 0 Hz hence this process can be done gradually after you have hit the cut-off frequency point reducing the frequencies below 20 Hz gradually hence the smaller the amplitude and gain is expected to be for the below 20 Hz signals and making sure all 0 Hz frequencies are cut-off. Hence, we might as well use first order filter as it requires less components and is easier to build than a second order filter and either one does the job correctly of removing the 0 Hz voltage offset. From research earlier, our smallest number in the EMG dominant range is 20 Hz hence cut-off frequency for the high pass filter is 20 Hz attempting to remove all frequencies below 20 Hz.



This graph shows the behavior of the first order high pass filter, after the cut-off frequency the voltage output decreases significantly hence decreasing the gain at frequencies below this lowering their amplitude. The frequencies below this are cut off at an exponential rate minimizing their significance, and at 0 Hz we see the amplitude is zero hence the DC offset should successfully be fully cut off with no interference theoretically.

Cut-off frequency = 20Hz

$$f_c = \frac{1}{2\pi RC}$$

$$20 = \frac{1}{2\pi RC}$$

Using a capacitor of 10 micro farads

[Equation]

Using a resistor of 1K ohms

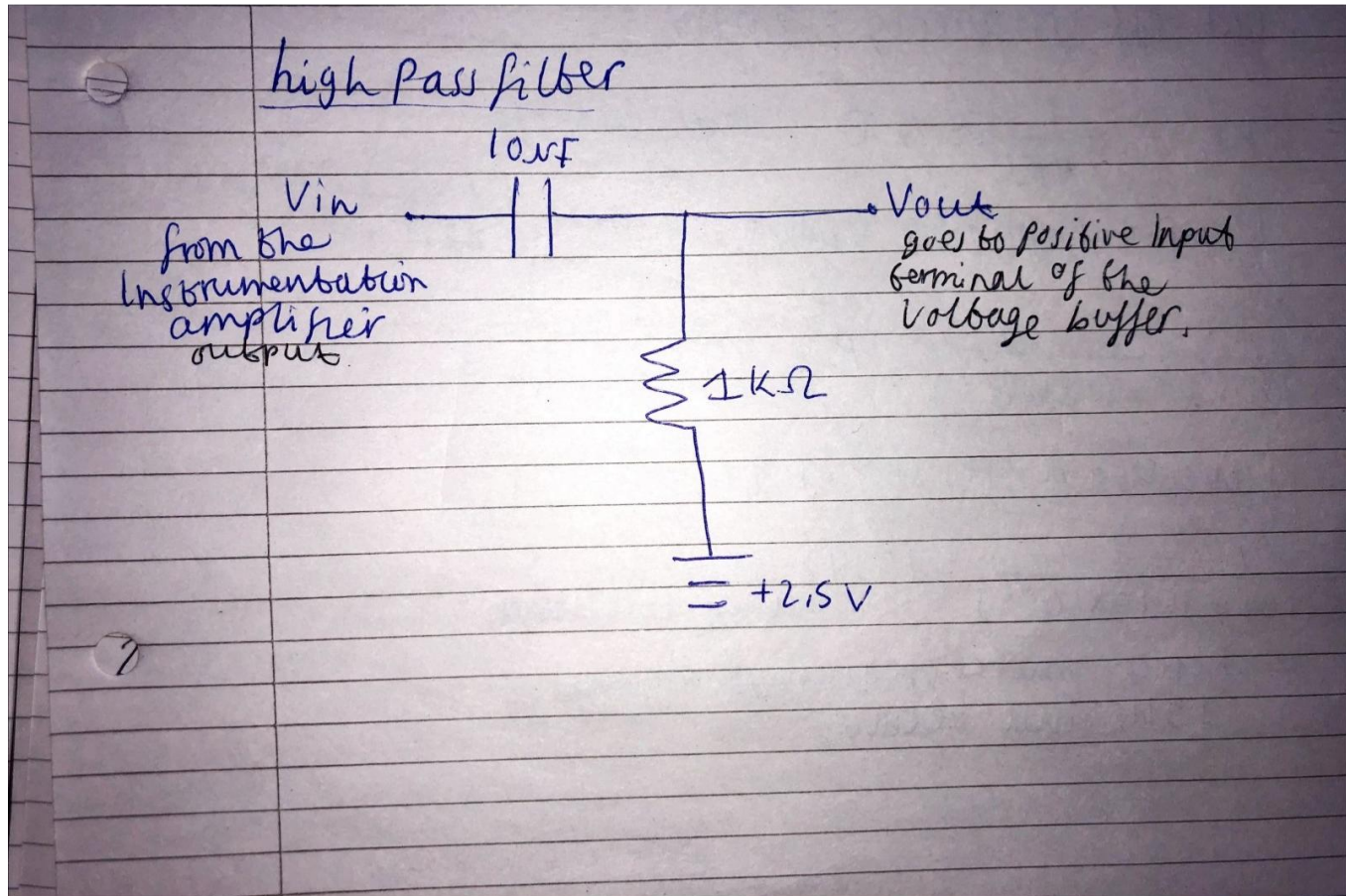
[Equation]

To summarize:

15 Hz is our actual cut-off frequency for the high pass filter

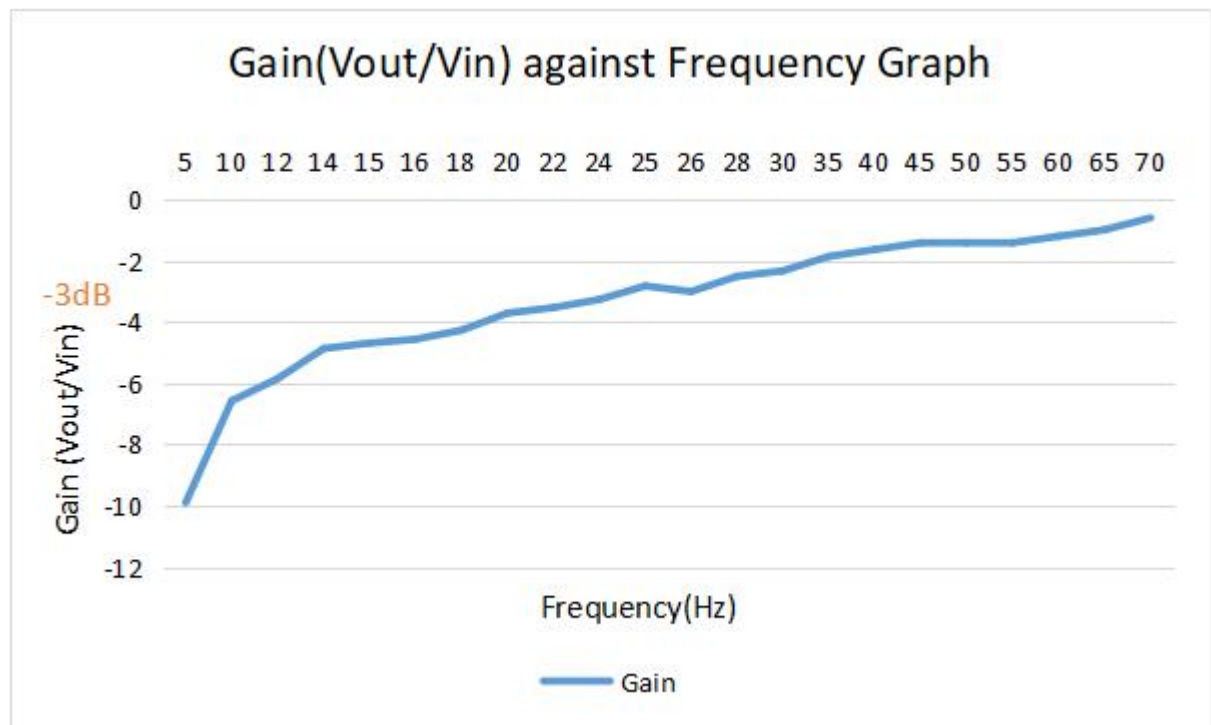
1k ohms resistor

10 micro farads resistor



Hence all frequencies around 15 Hz and less will gradually be removed, the smaller the frequency the lower the amplitude becomes hence by the time you reach 0 Hz the amplitude/gain is zero therefore all 0Hz frequencies are completely cut-off fulfilling the role of this high pass filter.

Characterization of high pass filter



Assigning the signal generator as an input at channel 1, setting a constant input voltage and channel 2 as the output of the circuit. As the frequency increases, the gain of the circuit increases meaning more output signal is allowed at higher frequencies in comparison to lower frequencies which proves the characteristics of a high pass filters. At around 15Hz the gain decreases dramatically reducing the amplitude / output voltages gradually for all frequencies below.
(See Appendix for tabled data of high pass filter.)

Non-inverting amplifier

Initially, it was used in order to finish off the amplification to around 5000 but then we decided to increase the instrumentation amplifier gain to around 20. This amplifier was used as it is non-inverting hence does not invert the signal at the output keeping it positive. Hence instead of using 500 as originally planned, we thought we'd try a smaller value around 200 as we increased the instrumentation amplifier gain, to test it out only as we know the op-amp is not acting ideally. The gain formulae for the non-inverting amplifier:

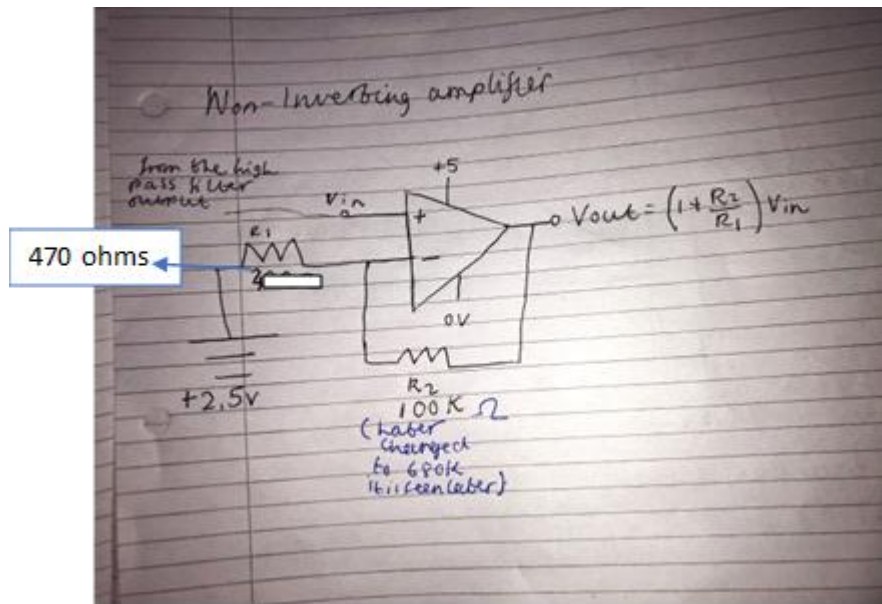
[Equation]

Using $R_2 = 100\text{K ohms}$

Using $R_1 = 470\text{ ohms}$

[Equation]

Hence the real overall gain is $22 \times 214 = 4708$ with later changes added if required.



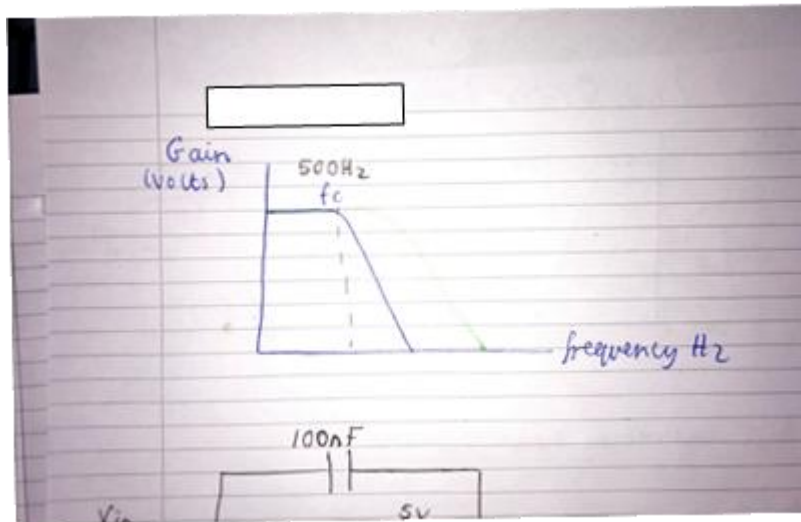
The value for R_2 was changed later as this part of the circuit was used to provide an amplification until we reached a desired amplification seen on the serial plotter without inverting the signal.

2nd order low pass filter

Our range was 20-150 Hz hence the frequencies we want to cut-off are all above 150 Hz this means that the cut-off frequency of the low pass filter would be 150 Hz this was the dominant frequency range but however we now decided to use the broader range that goes up to 500 Hz for convenience as if the broader muscle range is used then the same circuit design can be used for each muscle making the circuit more likely to fit in other muscles frequencies ranges. this also increase the like hood of it working as we are providing more frequencies as anomalies can lead to the signal frequency being out of the dominant range and therefore better to use the more normal broad range which is around "20-500" Hz to cut off noise and interference from other signals that will occur at high frequencies that are not in the EMG range. A 2nd order low pass filter will cut off the frequency values above 150 Hz at a much faster speed than a first order low pass filter getting rid of more unwanted frequencies reducing their amplitudes more quickly hence a lot less of the unwanted frequencies are present as they are removed a lot faster so it would be better to use a 2nd order low pass filter to get rid of the frequencies above 500 Hz as they work more accurately in removing the unwanted frequencies as soon as possible to prevent as much noise interference as possible in order to get a clearer signal removing the unwanted signals that are above the EMG range.

Hence our frequency range is now 20-500 Hz so we have more chance of including different muscle EMG signals that may not be in the dominant range. Also, a broader range means the same circuit can be successfully used for more groups of muscles.

Second order low pass filter graph



Here is a second order low pass filter graph which acts exactly like a first order filter in terms of cutting off frequencies, but this time aims to cut off all high frequencies above the threshold value. If you look at the green faint line this portrays a first order low pass filter, compare this to the blue clear one which is a second order filter hence the second order filter cut-offs unwanted frequencies more rapidly and reducing their amplitudes so there are less unwanted frequencies present therefore does a better job of cutting off all unwanted frequencies after 500 Hz keeping our frequency range in the desired range better than a first order filter would this is why this was used.

Cut-off frequency = 500 Hz

$$F_c = 1 \div (2\pi RC)$$

$$500 = 1 \div (2\pi RC)$$

Let's use a small capacitance as it is better and reduces noise caused by the components so capacitance of 100 nano-farads

[Equation]

[Equation]

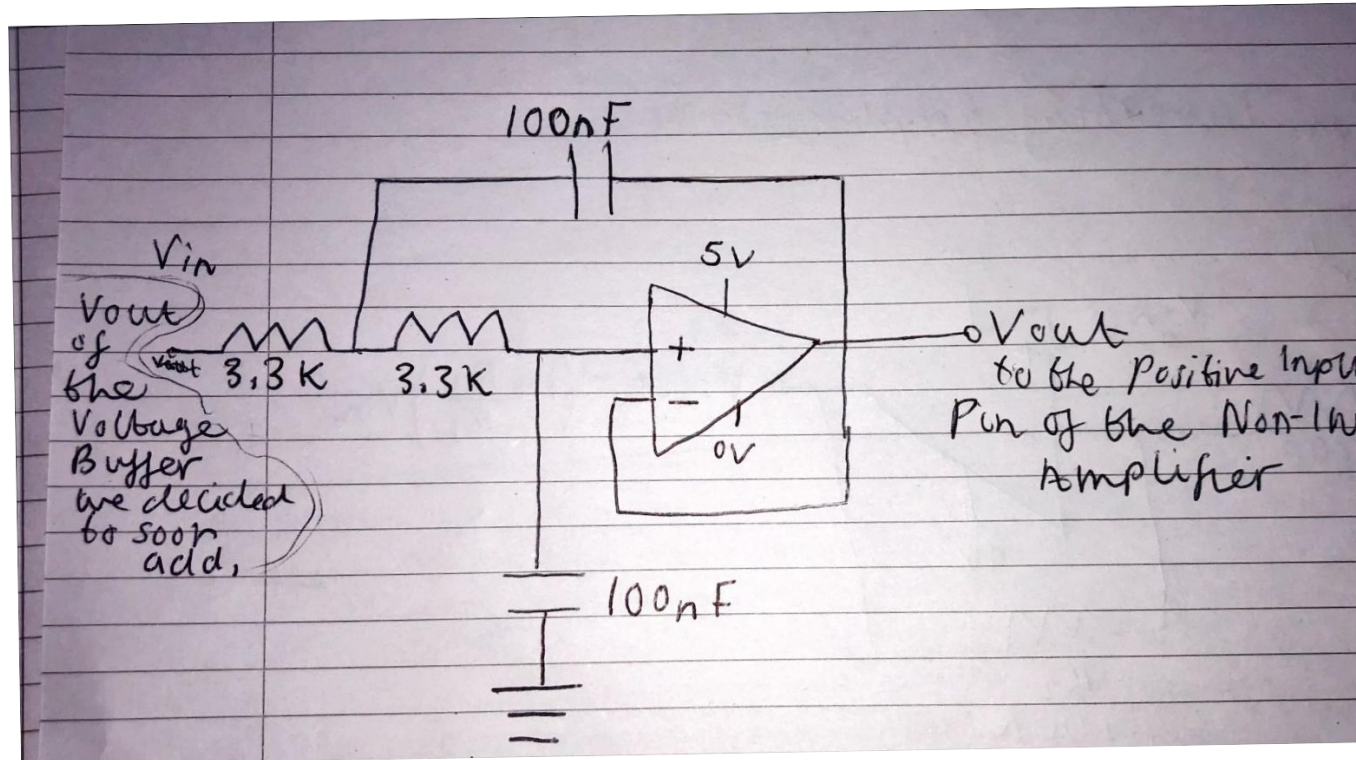
Using a resistance of 3.3K ohms

To summarize:

Cut-off Frequency: 482 Hz

The 2 Resistors = 3.3 K ohms

The 2 Capacitors = 100 nano-farads



Characterizations of low pass filter

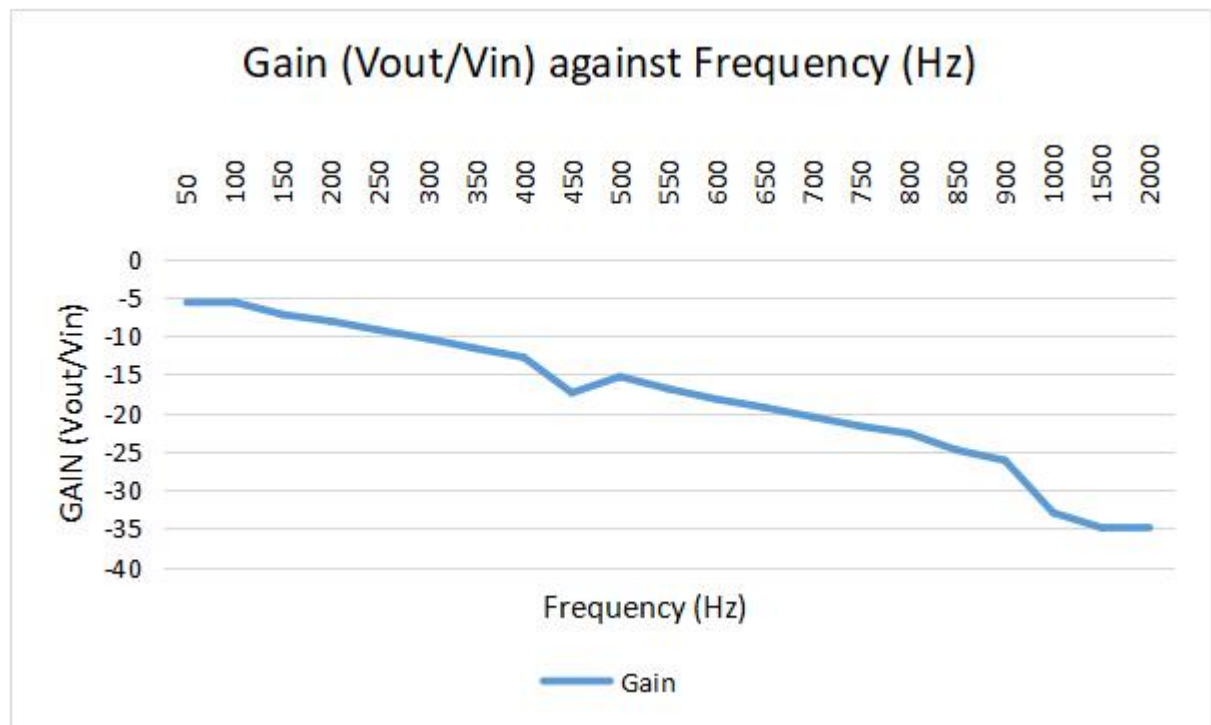


Figure : The Graph of Gain against Frequency of 2nd Order Low Pass Filter

As for the 2nd order low pass filter, the gain is high as it starts off with low frequencies. When the frequency increases, the gain slowly decreases relative to the frequency. This proves that signals above the cut-off frequency start to be filtered by the filter circuit.

Hence our new exact frequency range for our EMG using real values is 15-482 Hz.

Wednesday 14th December

Testing the click

We then tested with the oscilloscope to see if we got a sufficient gain, the input was set in the range of millivolts generated by the signal generator, an observation was made to determine whether the output of the signal on the oscilloscope was amplified, it wasn't amplified enough the amplification was very low, so we increased the gain of the non-inverting amplifier instead:

We replaced the R2 resistor for 680K ohms in order to add extra amplification:

$$1 + (680k / 470) = 1448$$

Hence our new overall gain is now $1448 * 22 = 31,856$

This did in fact amplify it to a decent level and the circuit testing was a success. It was then tested out using the Arduino Leonardo. where the code was run.

A problem occurred where the Arduino Leonardo was unable to be detected by any group members' laptops, this may be due to the conditioning circuit drawing out too much current hence we had to do a full circuit analysis to figure out why this was occurring and as a result of hours going by with no solution we decided to rebuild the entire circuit the next day with two buffers instead of one buffer as it helps to prevent

loading and to reduce current being drawn, and the result was a success as the Leonardo board was detected and the output was then observed using the serial plotter. The coding used for this testing is in the appendix.

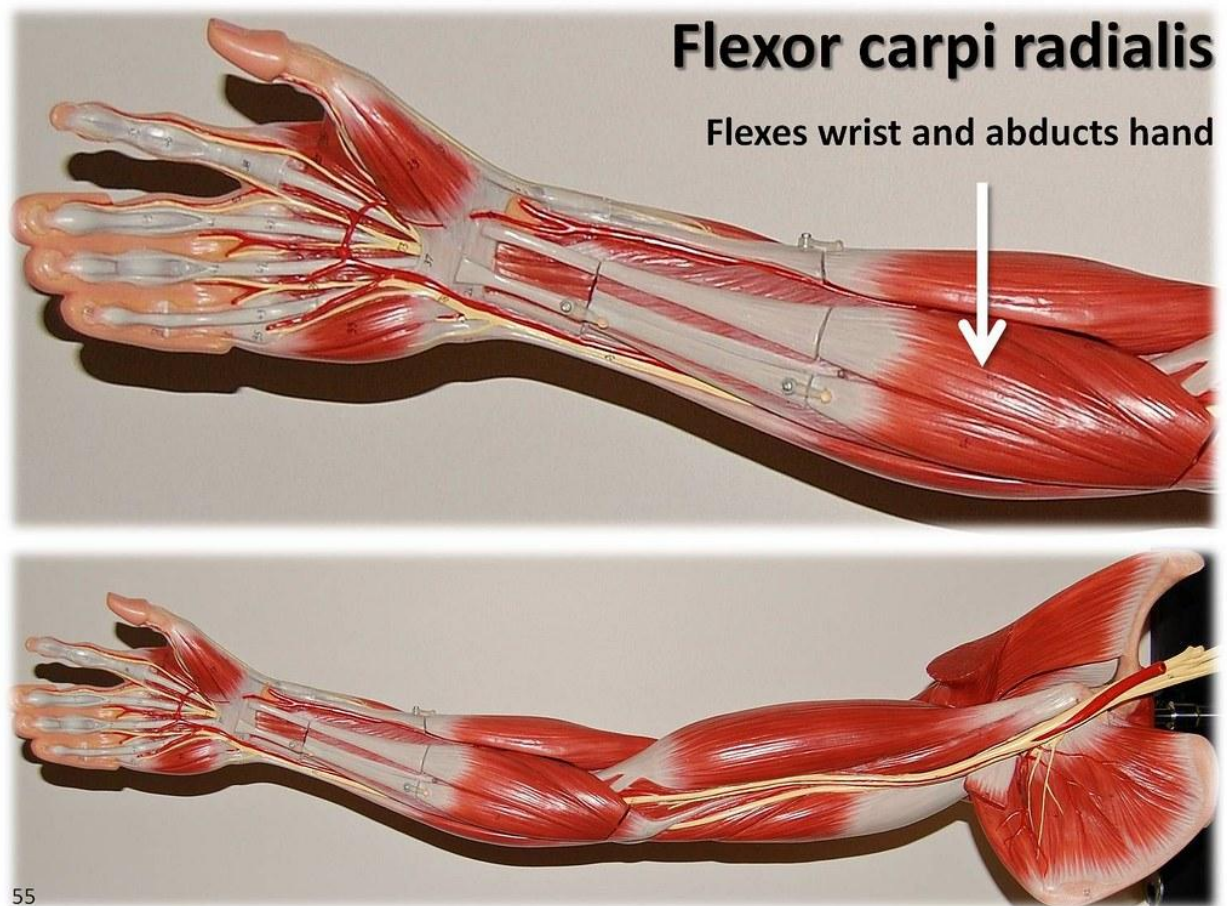
The 3 electrodes were then attached, as previously mentioned voltage reference was stuck to the elbow bone where there was the least muscle interference hence was used as a reference, the left and right electrodes were placed specifically on a superficial muscle (surface muscle) in a bipolar configuration and the distance between them should be between 1-2 cm apart.

Reference : <https://www.intechopen.com/chapters/40131>

The chosen muscle was the “flexor carpi radialis” which is a surface muscle in the forearm. This muscle is the long superficial muscle of the forearm and is in the first layer hence is a surface muscle closer to the wrist area hence was a valid muscle to test out for the click produced by an arm muscle.

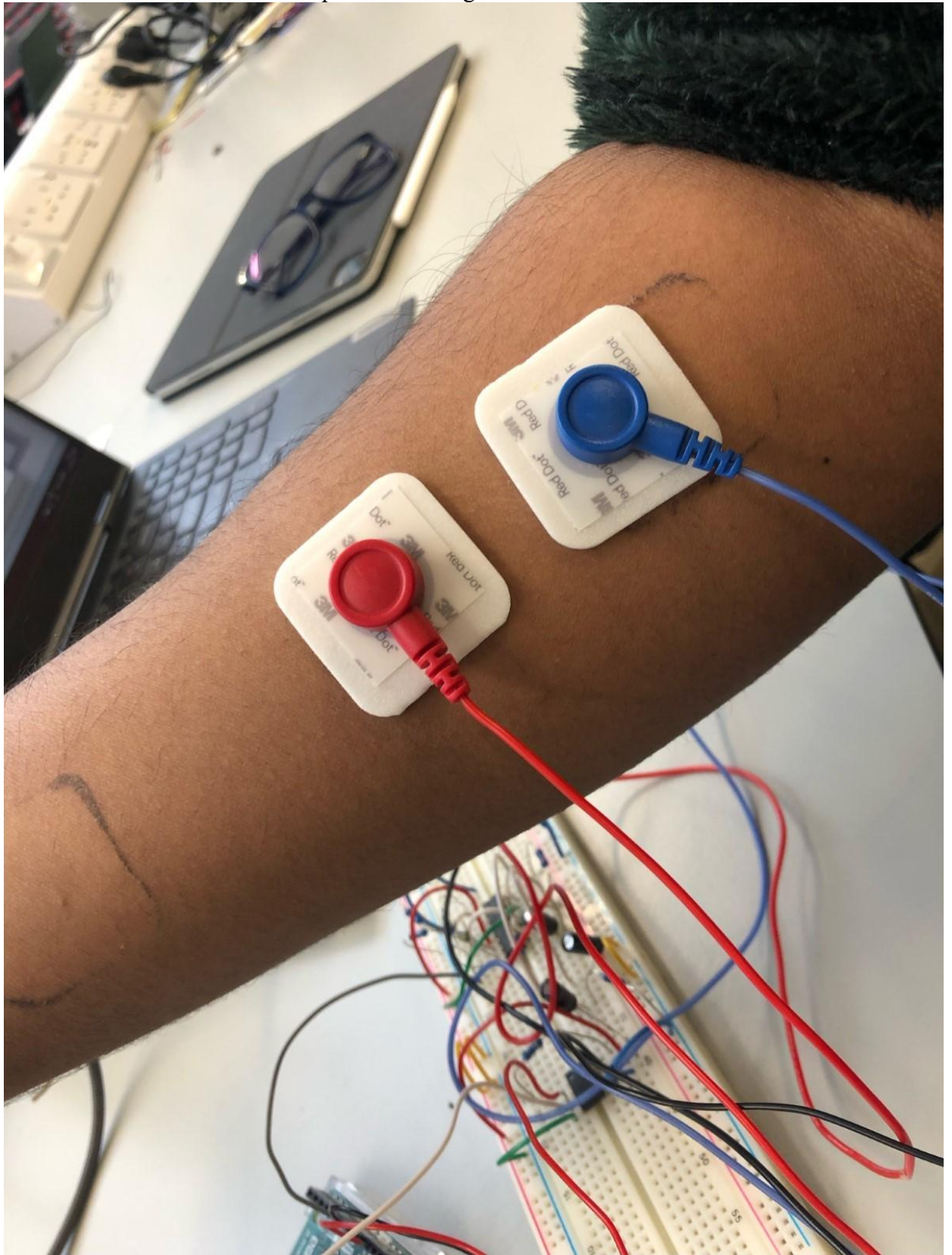
[flexor carpi radialis - Search \(bing.com\)](#)

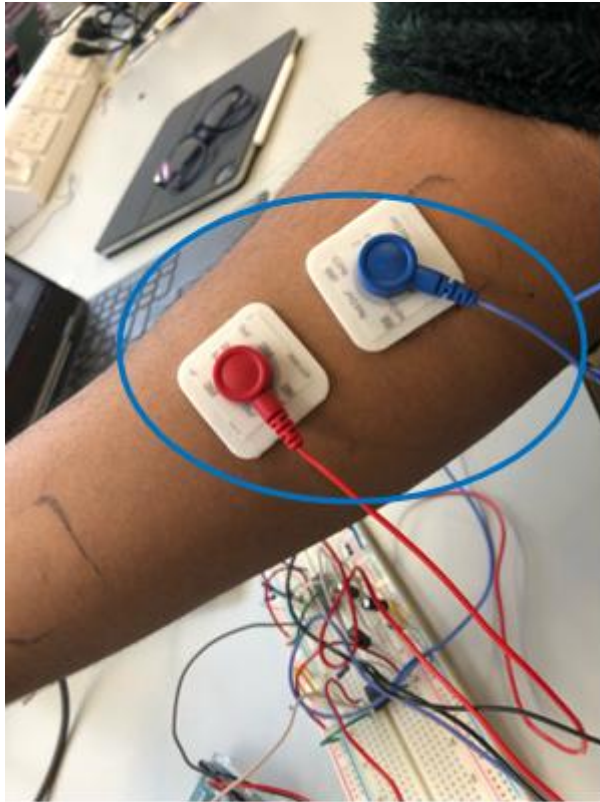
by moving it/ clenching multiple times we found the belly of the muscle and put the two electrodes close to each other either side of the muscle belly where the movement was felt. We placed the positive and negative electrodes close together in order to obtain a better EMG signal that is more concentrated.



[flexor carpi radialis - Bing images](#)

Electrode set up for the testing of the click circuit:





flexor carpi
radialis



Image 1: Electrode Placement on the User's Body

The left and right electrode were placed close together on the user's muscle with the black electrode as the reference point stuck on the elbow.

The positive and negative pins were the input to the instrumentation amplifier, the 0 volts in the circuit was connected to the 0 volts on the Arduino Leonardo board and the 5 volts in the circuit breadboard was also connected to the 5 volts on the Arduino board, the A0 was our input pin into the Arduino software, it was connected to the output of the signal conditioning circuit so the output of the non-inverting amplifier (pin 7) . This process would be repeated for all channel 1s whilst the next channels would just connect to the channel 1s breadboards 0,5 and the 2.5 (Vref) volts using wire connections and breadboards (as there is limited number of desired Arduino Pins to input into the board but this process still connects the channels to the Arduino board)

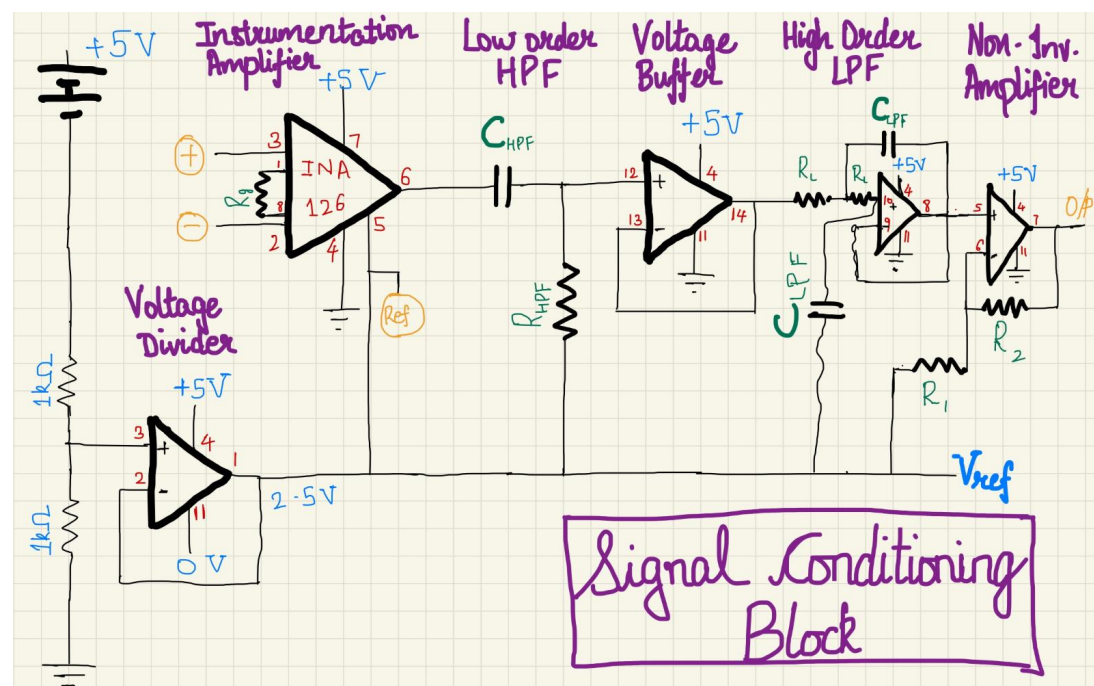
We then uploaded the serial plotter and the user clenched, every time the user relaxed the muscle, the base line was 240 in digital levels. This was when the muscle was

relaxed and then every time the user clenched the digital level went up to around 850.

Every time there was a clench the more of the muscle fibers from the flexor carpi radialis muscle were being activated hence the muscle is sending out more electrical pulses due to the muscle activation and the digital level increased majorly. The digital values correspond to voltage.

This was then used as a mouse to play the game. Every muscle clench corresponded to a click on the screen.

The clicking code can be seen within the appendix and is explained more later. Overall final circuit diagram for muscle signal conditioning:



The Instrumentation Amplifier, Low Order High Pass Filter, the Voltage Buffer, High Order Low Pass Filter, and the Non-Inverting Amplifier form the signal conditioning block. This block was planned to have been reconstructed for each individual channel and the output of the voltage divider from one circuit could be connected to the rest for offset purposes.

Instrumentation amplifier:

$R_G = 4.7K \text{ Ohms}$

High pass filter:

$C_{HPF} = 10 \text{ Micro-farads}$

$R_{HPF} = 1K \text{ Ohms}$

Low pass filter:

$R_L = 3.3 K \text{ Ohms}$

$C_{LPF} = 100 \text{ Nano-farads}$

Non-inverting amplifier:

$R2 = 680K \text{ Ohms}$

$R1 = 470 \text{ Ohms}$

To first set the V-ref to 2.5 volts the voltage divider circuit was used so that negative parts of the EMG don't get cut out. This was followed by a voltage buffer circuit to prevent loading.

Then the instrumentation amplifier takes the difference across the two input electrodes on the surface muscle and amplifies them in order to make them a more reasonable size.

Followed by a first order high pass filter to then remove the DC offset (at 0 Hz) before we fully amplify the circuit to get it into the 0-5 volts which was the Arduino power supply range.

This was then followed by another buffer to help prevent more loading and current being drawn out due to problems occurring with the Arduino not being able to detect our Arduino Leonardo board.

Then a 2nd order low pass filter to remove all frequencies above the range of 500 Hz as they are unwanted/undesired signals not within the EMG range hence causes noise interference in our signal giving us an inaccurate EMG so needed to be removed.

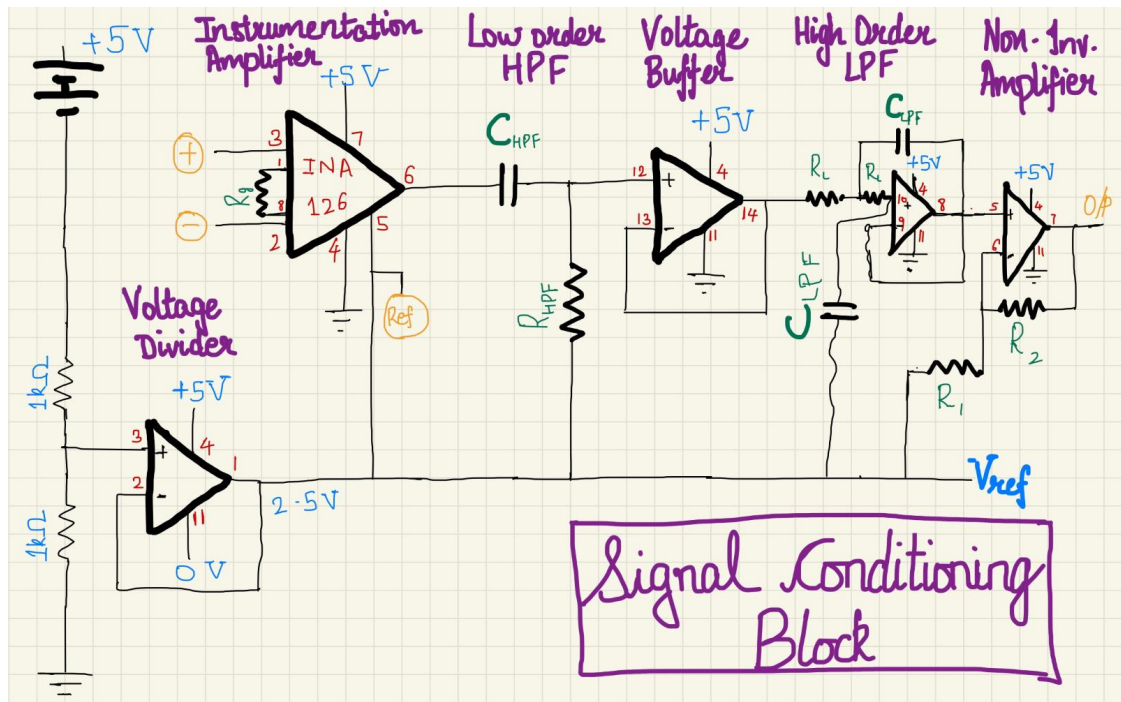
Then finally a non-inverting amplifier was used to increase the final gain so that the signal is displayed clearly on the serial plotter within the Arduino power supply range.

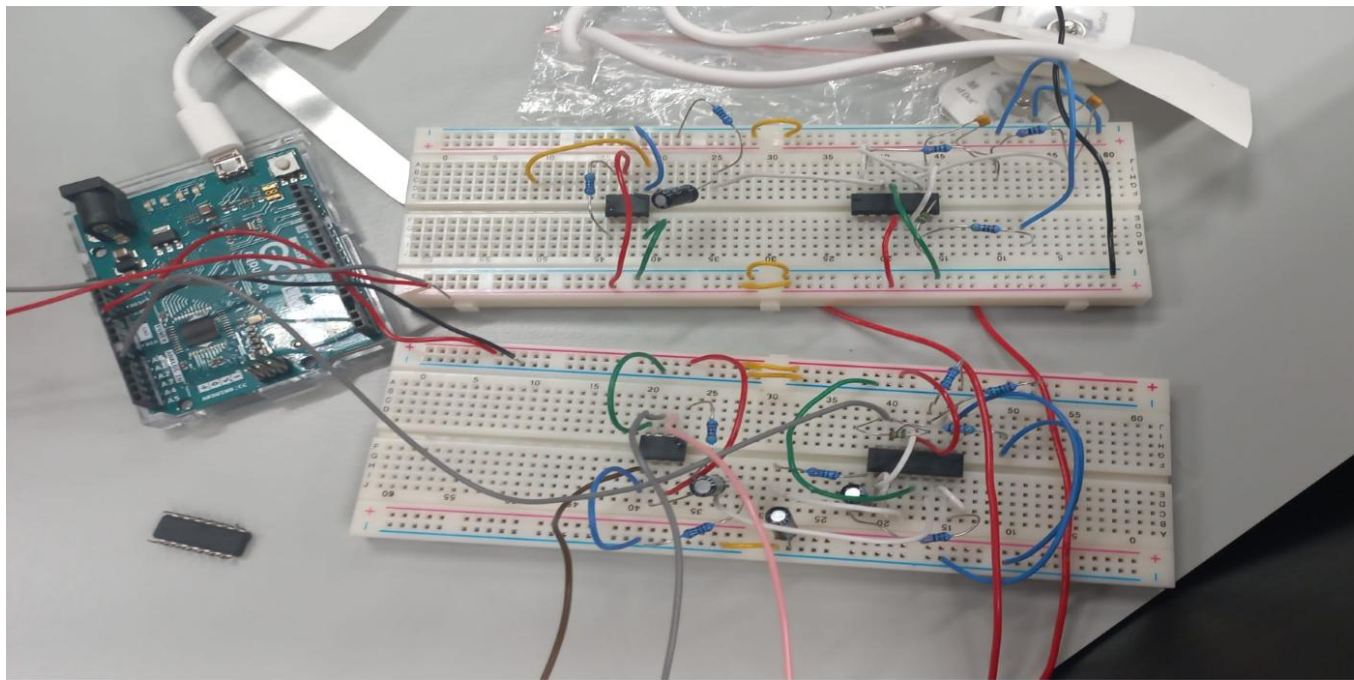
For the click code, we referenced the Analog Read example included in the Arduino IDE as it is efficient, well-structured and allows easy input of analog data within the range. The mouse module was included in the code and the click function was added within an if condition (if the analog read went past a certain output/ reached a certain digital level the mouse click. This is because the increase in digital level corresponds to the muscle clench. This led to a lot of misclicks between the click and the unflexing of the muscle. Hence as a result a delay was added shortly after to mitigate the misclicks and a time-controlled, misclick-free click function was designed.

Thursday 15th December

As the circuit designed above was a success, we then decided to use it for a second channel for now we kept everything the same for example the gain values and frequency values as we did use broader frequency range hence the frequency ranges of other muscles should lay within the range with a suitable gain used for all muscle types as well as they all have a similar voltage range. These two channels were for two different forearm muscles used to move the cursor left and right.

We rebuilt the circuit shown above already:





We didn't however rebuild the voltage divider & buffer circuit to go with it as we just connected our new circuit with a wire to the V_{ref} point of our original circuit hence this would also supply 2.5 volts as well as we connected it to the voltage reference of the first circuit. We then connected the 0 volts to the same vertical interconnected column on the breadboard that supplied 0 volts in the first circuit using a wire, same was done for 5 volts hence this way it was all connected to the Arduino Leonardo board.

Now we have 2 channels connected the purpose of these channels was for one to move the cursor left and one to move the cursor right by the activation of the 2 chosen arm muscles, hence one channel per muscle activated, each channel using 2 electrodes and then the V ref electrode remaining in the elbow.

Channel 1 had three electrodes used, one positive and one negative for the muscle, and a third electrode as a reference on the elbow.

Channel 2 had two electrodes used one positive and one negative on the specific muscle, the v ref electrode did not need to be used as it was done with channel 1 and the channel 1 circuit had connected the voltage reference to the channel 2 circuit already using the breadboard.

This meant two channels and 5 electrodes were used in total.

Our first problem faced was the fact we chose the bicep brachii and triceps as our two muscles for left and right movement by expanding and contracting the arm in order to move the cursor either left or right. However, when we tested the bicep brachii muscle alone contracting the upper arm, there was noise interference from the triceps in the signal due to their opposite and equal antagonistic contractions hence we decided to use two different muscles unaffected by each other to reduce noise interference from the other one.

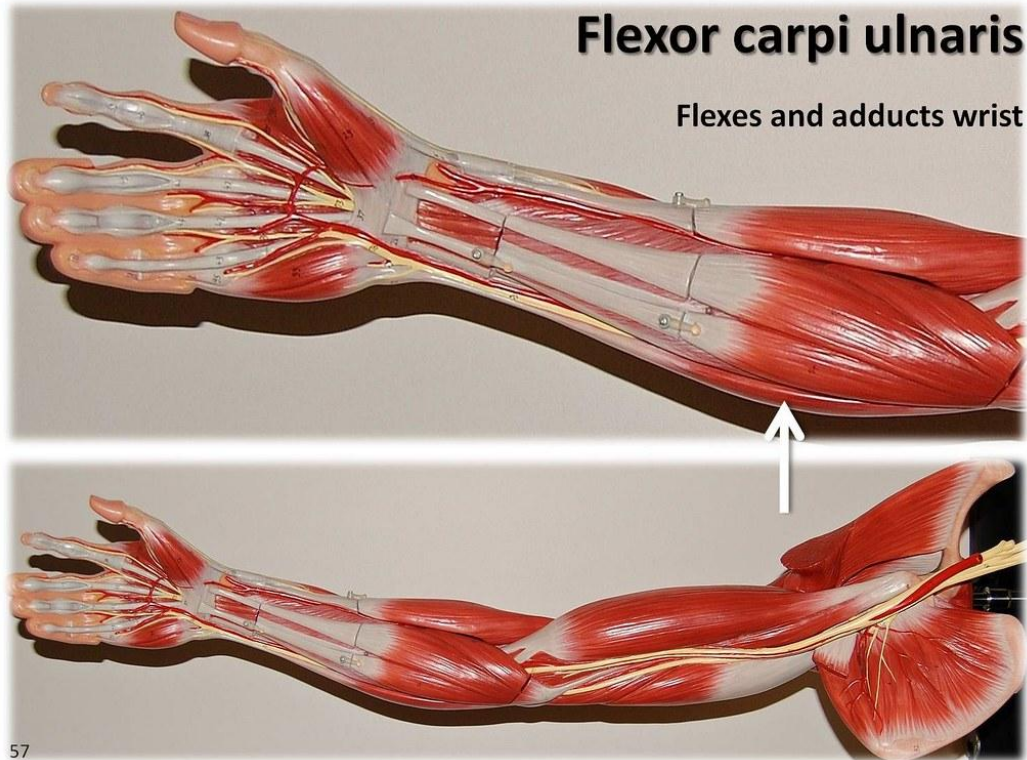
After researching for more surface arm muscles, we decided to use “flexor carpi ulnaris” for left movement of the mouse, which was activated when clenching the forearm left, hence activating the muscle. And the “brachioradialis” muscle for the right movement of the mouse which occurs when you clench forearm to the right using this muscle.

The flexor carpi ulnaris muscle is a thin, long muscle in the forearm it is considered a superficial muscle of the forearm as it is directly beneath the skin hence is a good muscle to use for surface electrodes as it is a surface muscle and can be detected was an EMG using skin-based electrodes.

[Flexor Carpi Ulnaris: Definition, Location & Function | Study.com](#)

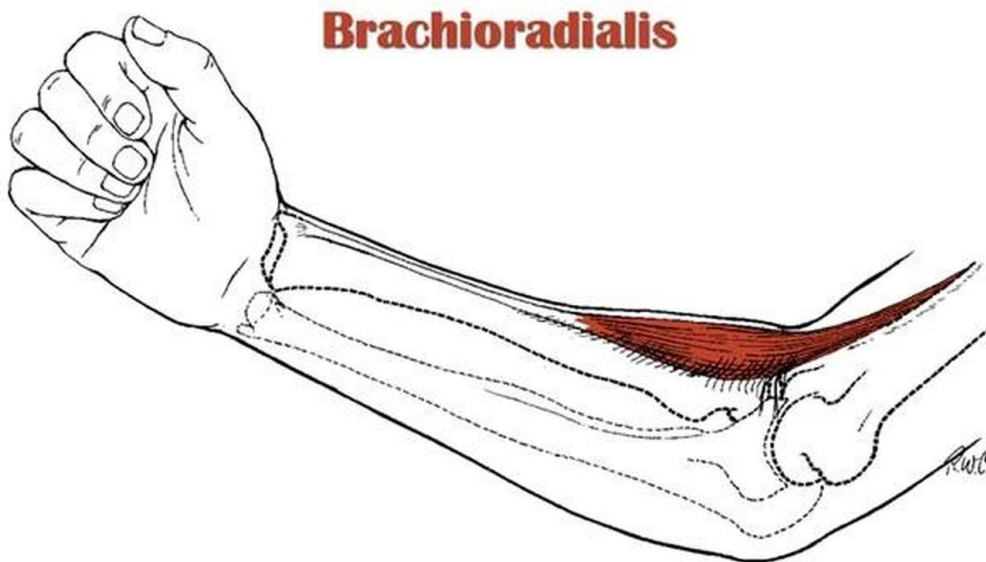
Brachioradialis is a “fusiform muscle located in the lateral part of the posterior forearm , it comprises the radial group of forearm muscles, which belong to the superficial layer of posterior forearm muscles” hence is part of the outer surface layer of forearm muscles so is another suitable muscle to use for surface electrode detection of electrical activity.

[Elbow and forearm: Forearm muscles and bones anatomy | Kenhub](#)



[flexor carpi ulnaris - Bing images](#)

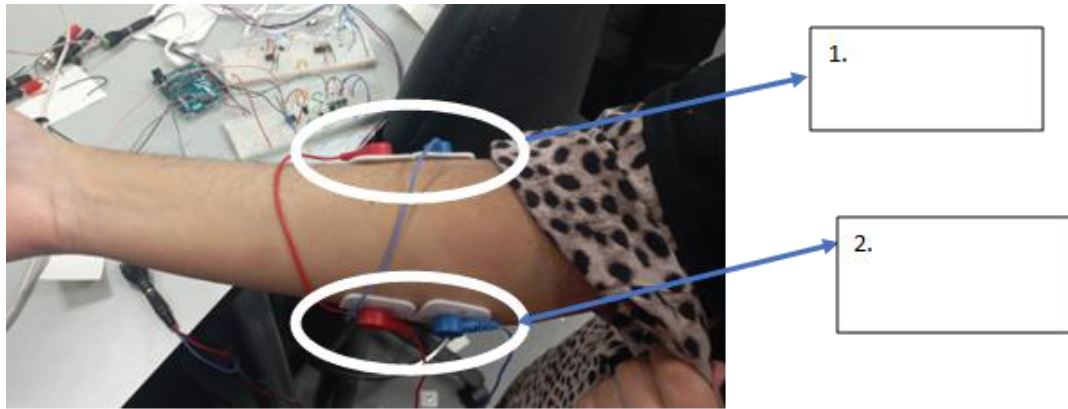
Muscle used to rotate the hand to the left to move the mouse to the left



Muscles used to rotate the hand to the right to move the mouse to the right.

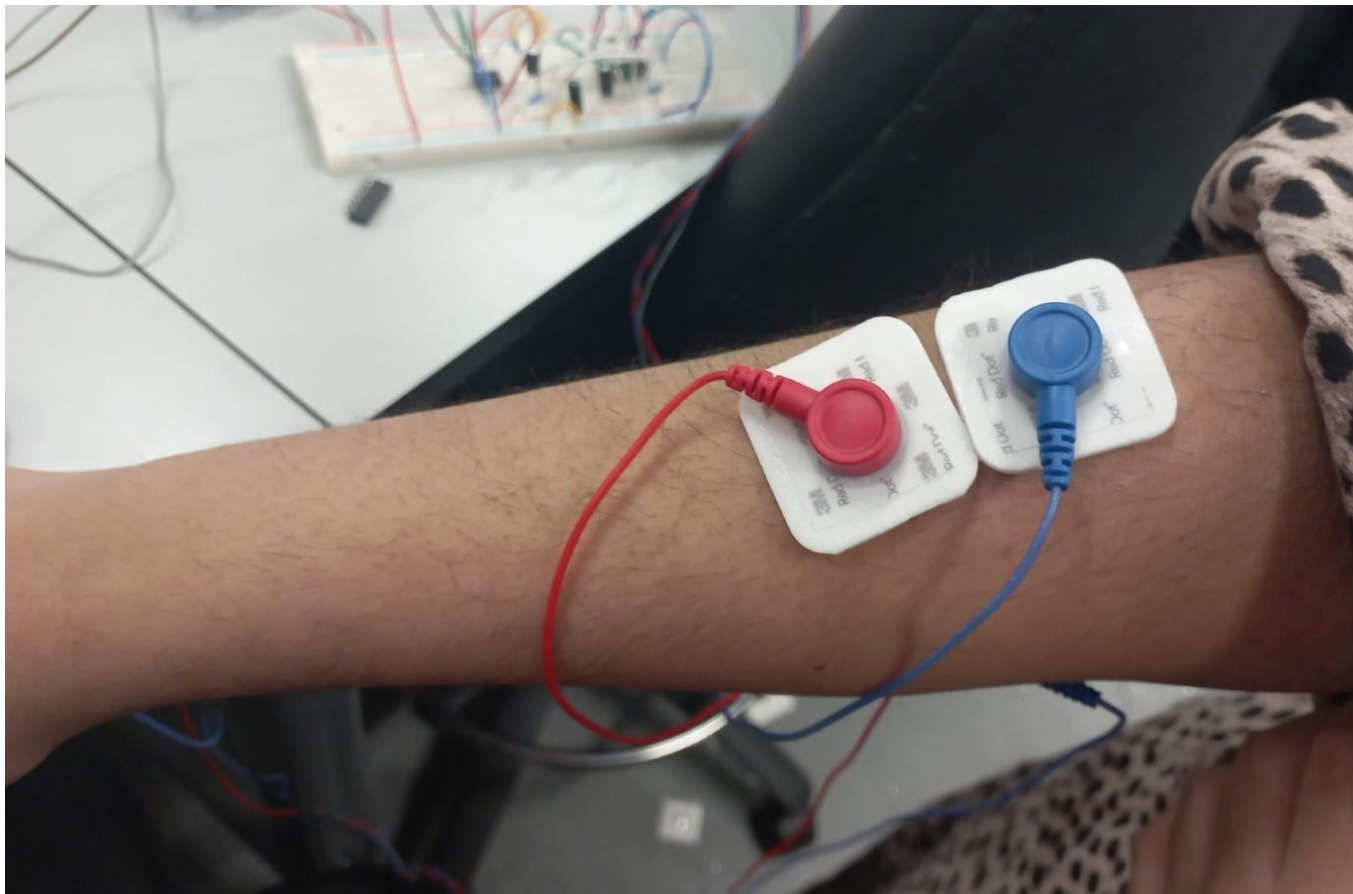
[brachioradialis - Bing images](#)

Electrode placement for the left and right cursor movement testing:

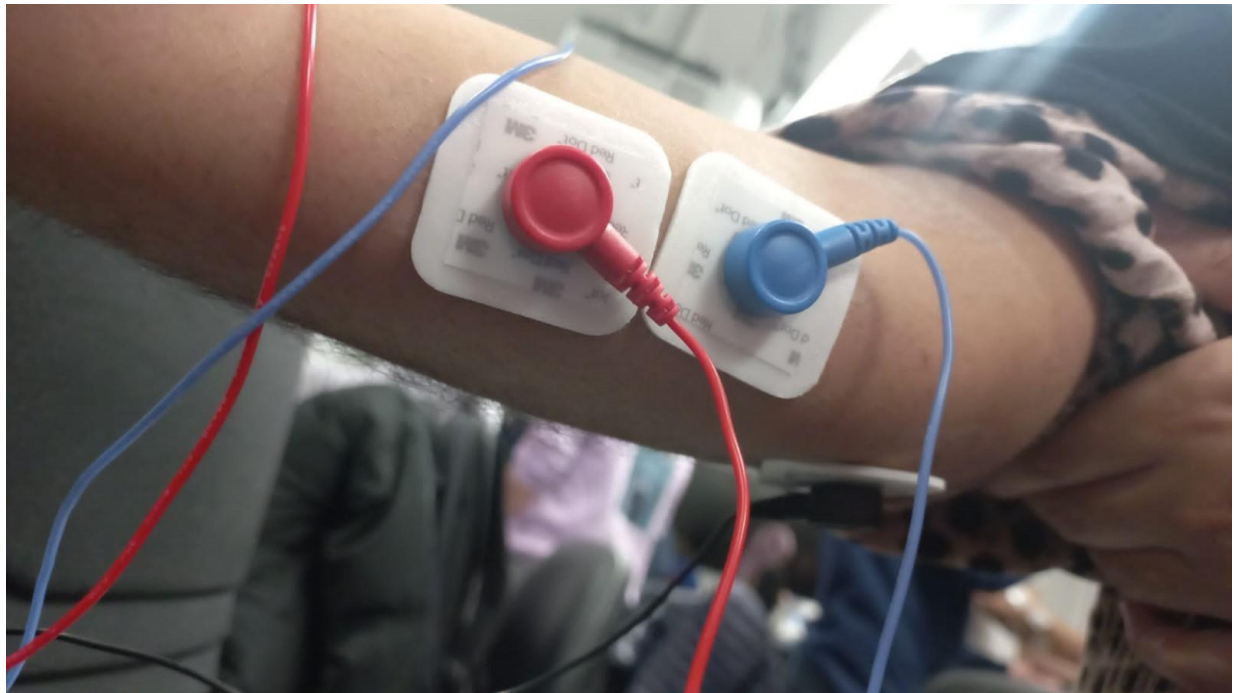


1)The electrodes placed close together on the “brachioradialis”

2)The electrodes placed close together on the “flexor carpi ulnaris”



2:



Here we placed the two sets of electrodes close together as it's where the muscle was strongly felt when clenching the forearm. Also, the clearer the EMG signal becomes the closer the electrodes are. As well as connected V ref to the elbow again as this was an undisturbed boney area.

We then formed the code in order to program the mouse to move left or right according to the activation of these muscles. (See appendix for code)

The code used for clicking was modified to now include two analog reads on different channels and hence, have two different if conditions for movement to the left and right. Based on the values seen on the serial plotter when the user was comfortably moving their muscles with minimum disturbance, the thresholds were set and the best values to comfortably move the mouse a reasonable distance were selected by trial-and-error and taking averages over multiple attempts in order to select the best threshold values to have the curser working as efficiently as possible.

The two different channels worked independently of each other hence the signal for each channel were tested one at a time, first starting with the left clench channel, every time there was a left clench we expected the digital level to significantly rise in order to pick a suitable threshold value for the flexor carpi ulnaris muscle clench to move the curser left. Then testing the right channel every time the brachioradialis muscle activated from a right clench we expected to see a major decrease in digital levels in order to help pick the best threshold value for each one.

We then tested the code with the serial plotter to see if the digital levels increased only for the muscle activations of the two selected muscles attached to the electrodes. One problem we faced was saturation in the signal, we saw the signal was saturated at first cutting parts of the signal off, so we checked our circuit resistor and

capacitor values to see if the gain was the same in the second circuit as the first, in order to test resistor values we used an online resistor color band chart and saw for the non-inverting amplifier we accidentally used 200 ohms for our R1 instead of 470 ohms hence this would have led to a greater amplification which led to saturation of the signal, as a result of this human error we then changed it to 470 ohms and the saturation problem was fixed.

Our second problem was that the electrodes were placed on an overweight group member on a rather fat surface muscle and so, the fat layers under the skin made it harder for the skin surface electrodes to detect the muscle activation during muscle clenching, especially for the brachioradialis muscle causing the right mouse movement to work insufficiently.

In order to fix this problem, we placed the electrodes on a relatively healthier group member (based on the BMI Index) instead and both muscle movements were detected and hence successfully moved the mouse. This included constant threshold adjustments in the code until we felt the mouse was moving the most effectively. The different threshold values for the muscles were due to how much the muscle was activated which led to the strength of the user's muscles. We experimented this by plotting the results on the serial plotter then getting the electrode wearer to actively clenching their forearm left and right comfortably and seeing the digital levels for each comfortable undisturbed clench, then take an average of this digital levels and use it to set the code thresholds for both muscles. This way the mouse could move left or right more effectively.

As it then successfully worked, we played the game moving the Christmas tree left and right by activating the desired muscles from rotating the hand!
(See appendix for links to previous coding attempts in GitHub)

What Was Unable To Be Achieved

There were some aspects of the scenario that we were not able to achieve due to time restrictions. The upwards and downwards movement was unfortunately unable to be tested upon given the short time left.

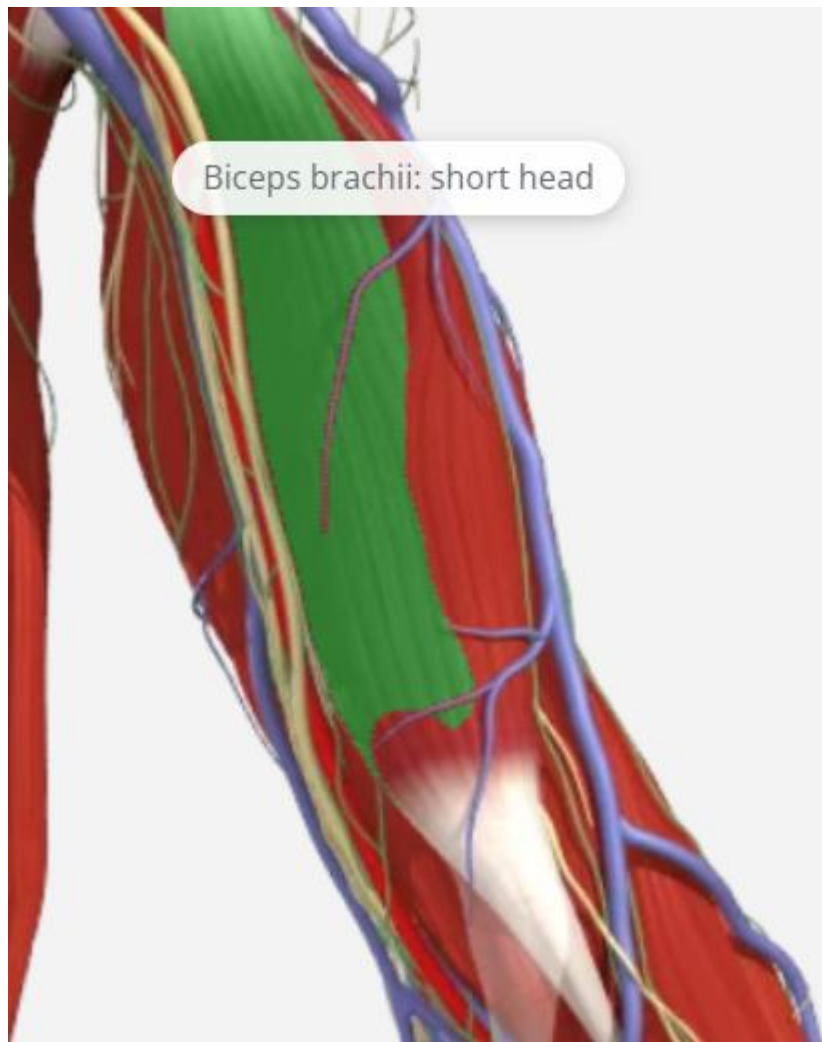
If however, given the time and opportunity, the upwards and downwards movement of the mouse cursor would have been carried out, from our research the following muscles would be used:

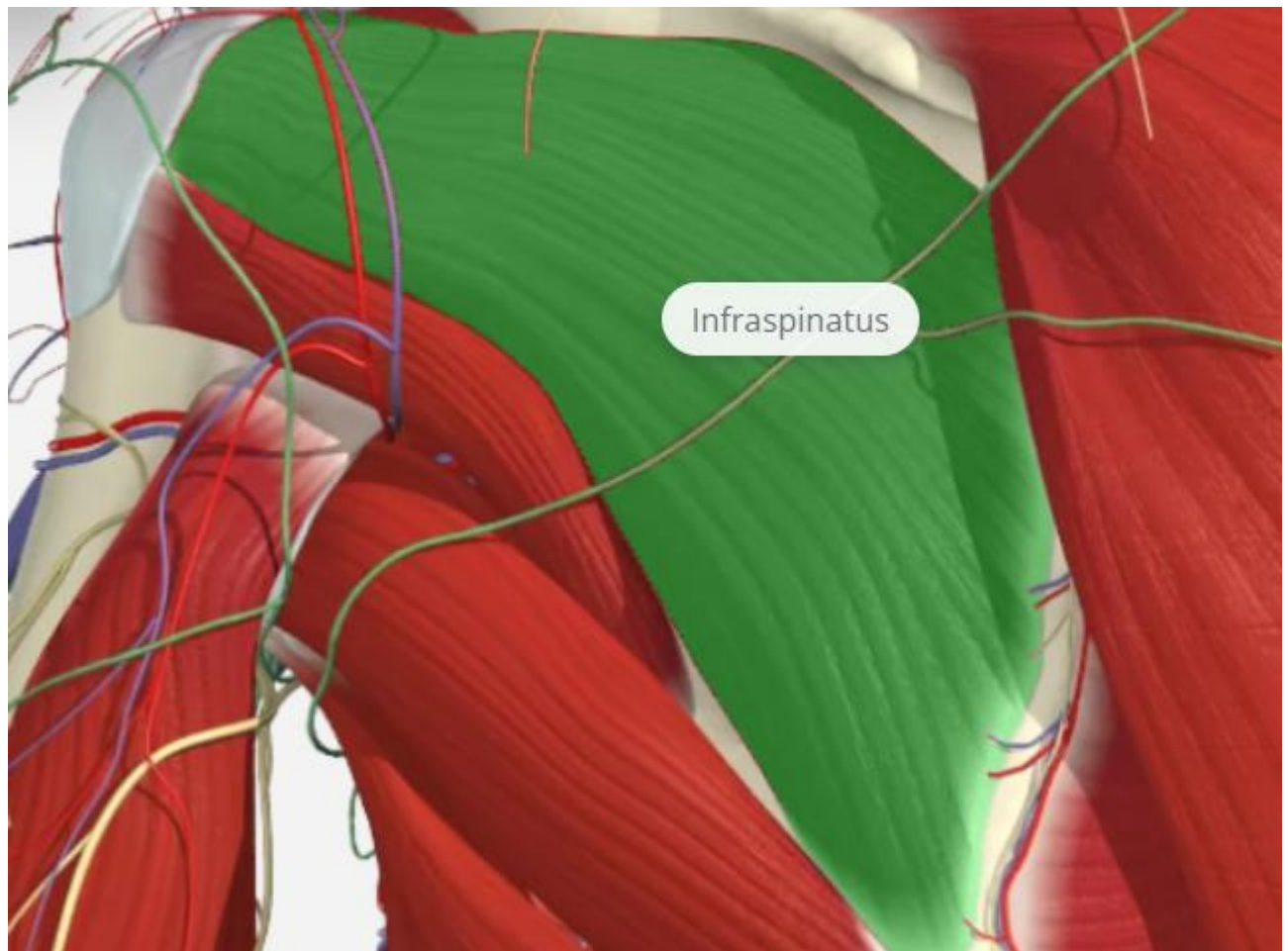
For Upwards Movement

Anterior arm muscle- Biceps Brachii (short head) top part of arm is contracting
Posterior shoulder muscle – Infraspinatus muscle is contracting

For Downwards Movement

Anterior arm muscle- Biceps brachii (short head) top part of the arm is extending
Posterior shoulder muscle – Infraspinatus below the arm is extending.





Images of the Biceps Brachii (short head) and the Infraspinatus obtained from anatomy.tv

[Anatomy.tv Titles](#)

The **biceps brachii muscle** is a two-headed muscle in the anterior compartment of the arm, its function is to flex at the elbow and supinates the forearm. As it is an exterior muscle it is hence a surface muscle is suitable for usage of electrode-based EMG with skin attached electrodes.

[Biceps brachii muscle | Radiology Reference Article | Radiopaedia.org](#)

We then decided to use the upper arm nearby shoulder muscle due to the previously explained interference problems between using the biceps and triceps at the same time as they act in accordance with each other. Hence, we decided to use the Infraspinatus muscle as the other main arm muscles had also already previously been used. “The **infraspinatus** is one of four muscles that make up the rotator cuff, which helps your arm and shoulder move and stay stable” it is the outer exterior upper arm muscle used for rotation and hence is therefore suitable to use a surface muscle for EMG detection via surface electrodes.

When the user wants the cursor to move upwards, the user can contract the biceps and the infraspinatus muscles and when the user wants the cursor to move downwards, the user can relax the biceps brachii and infraspinatus muscles.

[Infraspinatus Pain Causes, Symptoms, and Treatments \(healthline.com\)](#)

[Biceps brachii muscle | Radiology Reference Article | Radiopaedia.org](#)

This testing would undergo the same process as the left and right cursor testing with the two electrodes placed closer together on where the muscle is felt the most on the skin during expansion and contraction, each muscle programmed by a channel, V ref remaining on the elbow and testing with extending and contracting arm movements to then programmed to move the mouse up and down. Overall, 4 electrodes (2 for each muscle) and 2 channels with two analog reads on the 2 different channels. We would undergo the same threshold testing as previously mentioned in order to make sure the cursor works as effectively as possible. On the serial plotter we would expect an upwards increase in digital level for clenching the arm and a major decrease in digital levels for expanding the arm.

If this was a success, we would further test with all 5 channels together with 5 analog reads one for each channel, 2 electrodes for each muscle (5 muscles in total) one as the elbow reference. One channel for left mouse movement, one channel for the right mouse movement, one channel for the up movement, one channel for the down movement, one channel for the click, 11 electrodes in total combining these channels and their codes will program them to work all at once hence each feature can be used all at once which was our overall goal to provide an effective useful cursor for clients with no hands and limited arm movements to enable them to use computer devices.

APPENDIX A

Tabled Data for High Pass Filter

HIGH PASS FILTER			
Frequency (Hz)	Vin (V)	Vout (V)	Gain
5	10.0	3.2	-9.897000434
10	9.8	4.6	-6.56936488
12	9.8	5.0	-5.845121427
14	9.8	5.6	-4.860760974
15	9.6	5.6	-4.681664121
16	9.8	5.8	-4.555961643
18	9.8	6.0	-4.261496506
20	9.8	6.4	-3.700922034
22	9.6	6.4	-3.521825181
24	9.6	6.6	-3.25454595
25	9.4	6.8	-2.812378818
26	9.6	6.8	-2.995246407
28	9.6	7.2	-2.498774732
30	9.4	7.2	-2.315907143
35	9.4	7.6	-1.846285226
40	9.4	7.8	-1.620665018
45	9.4	8.0	-1.400757332
50	9.4	8.0	-1.400757332
55	9.4	8.0	-1.400757332
60	9.4	8.2	-1.186280024
65	9.4	8.4	-0.976971351
70	9.2	8.6	-0.585787522

Tabled Data for 2nd Order Low Pass Filter

2ND ORDER LOW PASS FILTER			
Frequency(Hz)	Vin(V)	Vout(V)	Gain
50	10	5.2	-5.679933127
100	10	5.2	-5.679933127
150	9.8	4.24	-7.277204382
200	9.8	3.84	-8.137897026
250	9.8	3.36	-9.297735966
300	9.8	2.96	-10.39868729
350	9.8	2.56	-11.65972221
400	9.8	2.24	-12.81956115
450	9.8	1.32	-17.41304289
500	9.8	1.68	-15.31833588
550	9.8	1.4	-16.9019608
600	9.8	1.2	-18.24089659
650	9.8	1.06	-19.31840421
700	9.8	0.92	-20.54876497
750	9.8	0.8	-21.76272177
800	9.8	0.72	-22.67787159
850	9.8	0.56	-24.86076097
900	9.8	0.48	-26.19969677
1000	9.8	0.22	-32.9760679
1500	9.8	0.176	-34.91426816
2000	9.8	0.18	-34.71907141

APPENDIX B Arduino C++ Code for Clicking

```
#include <Mouse.h>
// the setup routine runs once when you press reset:
void setup() {
  // initialize serial communication at 9600 bits per second:
  Serial.begin(9600);
}
```

```
// the loop routine runs over and over again forever:
void loop() {
  // read the input on analog pin 0:
  int sensorValue = analogRead(A0);
  // print out the value you read:
  Serial.println(sensorValue);
  if (sensorValue > 800) {
    Mouse.click(); //click motion
    delay(1000); //delay to avoid misclicks while unflexing
  }
}
```

```

    delay(10);          // delay in between reads for stability
}

```

APPENDIX C Two-Channel Arduino C++ Code

```

#include <Mouse.h>
// the setup routine runs once when you press reset:
void setup() {
    // initialize serial communication at 9600 bits per second:
    Serial.begin(9600);
}

// the loop routine runs over and over again forever:
void loop() {
    // read the input on analog pin 0:
    //int Bicep = analogRead(A0); //Used for thresholding
    //int Tricep = analogRead(A1); //Used for thresholding
    // print out the value you read:
    Serial.println(Bicep);
    //Serial.println(Tricep);
    if (Bicep > 350) {
        Mouse.move(-40,0);
        delay(50); //Higher delay because this movement was more sensitive
    }
    if (Tricep < 848) {
        Mouse.move(40,0);
        delay(10);
    }
    delay(10);          // delay in between reads for stability
}

```

APPENDIX D

Note: this code states “biceps” and “triceps” however we later did change these two muscles for left and right movement to two new muscles, biceps should theoretically be replaced with “flexor carpi ulnaris” for left movement and the triceps should be replaced with “brachioradialis” for right movement. However, these identifiers do not affect the functioning of the program hence the program worked effectively still unimpacted by this therefore it was not changed due to limited time to test our program but theoretically these would be the correct new labels. As portrayed, the threshold digital values were 848 and 350.

GitHub Coding Attempts

<https://github.com/Biomedical-Engineering-Undergrads/scenario-4-2022-group-12>